

Guidelines for Writing a Computer Science BSc Thesis and Information on the Final Examination

At the Faculty of Informatics, students of all study programmes are have compose a separate, independent bachelor thesis for each of their programmes. (See section 76 (3), Academic Regulations for Students, henceforth ARS.)

The submitted thesis work is required to be an original intellectual product, the use of referenced literature is required to do according to the general rules of copyright. In the case of thesis works the following acts are considered plagiarism:

- literal quotation without quotation marks and reference;
- citation of content without reference;
- presenting others' published thoughts as own thoughts.

For more details, please read ARS sections 74/A-C.

Title

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THESIS TOPIC REGISTRATION FORM

◆ [Essential guide - Thesis topic registration](#) ◆

Thesis Topic Registration Form has to be submitted through Neptun:

Administration / Requests / Available request forms / IK – Thesis Topic Registration Form (BSc)

A student can have his/her own topic, in case he/she finds a supervisor for it or can chose from the [list given by the Faculty](#). In column Training, look for CS BSc.

Students have to provide the following data:

- Student's data
- Study programme
- Supervisor's data
- Title of the thesis
- a max. 1 page long description of the topic so that the difficulty and complexity of the task undertaken by the students can be clearly seen and evaluated.

Before submitting the Thesis Topic Registration Form, please contact the supervisor to discuss the topic and to check his data to be provided in the form.

Supervisors can be lecturers and also researchers who are not employed by the University, who are required to obtain at least a level of MSc degree in the relevant field. In case the student has an external supervisor, the Faculty will appoint an internal supervisor for him/her.

The topic and the supervisor of the student's thesis will be approved by the head of the department responsible for the programme.

The thesis topic has to be chosen (via the Thesis topic registration form) at least 6 months before the Final Exam period. (ARS section 77 (4a)).

The Deadlines for submitting the Thesis Topic Registration Form are the following:

- **15 October** in case of taking the Final Examination in June
- **15 May** in case of taking the Final Examination in January

The extended deadlines for submitting the Thesis Topic Registration Form for an extra administrative fee (late fee) are the following:

- 16-29 October in case of Final Exam in June
- 16-29 May in case of Final Exam in January

The student can change the topic of the thesis up **until four months before the Final Examination**, but between changing the topic and the beginning of the final exam period at least 4 months has to pass. (ARS section 77 (8a)).

Deadlines for submitting IK-Request for changing the thesis topic

- 1 February in case of taking the Final Examination in June
- 31 August in case of taking the Final Examination in January

In case of inappropriate Thesis Topic Registration Form, the supervisor will send the request back for correction.

Students can choose their thesis topic by themselves, by contacting their previous lecturers inquiring whether he/she would accept their supervision.

The **IK-Request for changing the thesis topic can be submitted only if:**

- either the topic OR the supervisor has changed since the submission of the Thesis topic registration form. In case both – the topic AND the supervisor – has changed, the student has to submit another IK-Thesis topic registration form.
- an IK-Thesis topic registration form has been submitted and accepted beforehand

The **Thesis Topic Declaration** form is **valid for 2 years**, after which a new TTD request must be submitted (even if neither the topic nor the supervisor has changed).

THESIS CONSULTATION

The BSc in Computer Science students have to complete Thesis Consultation subject. It proves that they have regularly consulted with their supervisor and got to the point where their work is complete at least 60-70%.

BSc in Computer Science students have to register for the Thesis Consultation subject during the registration period in Neptun in case they intend to finish their studies (obtain absolutory/pre-degree certificate) in the current semester. The Thesis consultation subject is published in Neptun with the supervisor as lecturer.

You are required to following steps:

- Check whether the Thesis consultation subject is published and available in Neptun with a subject code indicated in your curriculum (check your curriculum) with your supervisor as lecturer.
- Register for the subject in case you intend to complete your studies (obtain absolutorium) in the current semester.
- In case the Thesis consultation subject is not available with the correct subject code with your supervisor as lecturer, please contact your supervisor, ask him/her to publish the subject as soon as possible.

The fulfillment of the Thesis Consultation subject is recorded in Canvas system. The Thesis consultation grade is administrated in Neptun by the supervisor.

Consultation has to fulfilled in order to obtain the required amount of credits of the Thesis Consultation subject and the Thesis itself.

Deadlines for Thesis Consultation

CONTENT OF THE THESIS

The thesis consists of an ICT solution (i.e. a program) for a practical problem. The complexity of the task(s) set in the thesis must not exceed the capabilities and skills of an average student; i.e. the task(s) can be completed in max. 600 working hours, worth 20 credits. If the volume of the task requires extra time and effort, several students can co-work to write the thesis together. In this case, it must be clarified which part of the thesis is which student's work.

The thesis must contain the following sections as independent chapters:

1. Introduction
2. User Documentation
3. Developer Documentation

The source code of the program must be enclosed.

The Introduction consists of a reason for choosing the given topic and a brief and plain description of the problem to be solved.

The User Documentation contains

- a short statement of the problem solved,
- a brief description of the methods used,
- and all the information required for using the program.

The **Developer Documentation** contains

- a detailed specification of the problem
- a detailed description of the methods used, a definition of the terms used,
- a description of the logical and physical structure of the program (data structures, databases, module resolution),
- a testing plan and test results.

FORMAL REQUIREMENTS OF THE THESIS

Physical requirements of the thesis:

- Paper: white, size A4
- Font size: 12 points

- Rows: line-adjusted, 1,5 line spacing
- Margin: inside: 2,5 cm, outside: 2,5 cm, bottom: 2,5 cm, top: 2,5 cm
- Page number: continuously from the contents to the bibliography. Marked with Arabic numbers.
- Main chapters of the thesis (Heading 1) are required to start on a new page.

Structural requirements of the thesis:

- The header of the title page of the thesis must bear the ELTE logo, the name of the university, the faculty and the supervisor's department, respectively. The main body of the same page is has to contain the title of the thesis, the candidate's name and his/her training programme, the supervisor's name, the external consultant's name, the place of submission and the year of the thesis defence.
- Table of contents
- The thesis has to contain all the three chapters prescribed as obligatory in Section II (Introduction, User Documentation and Developer Documentation).
- The thesis must be completed -- if the topic requires -- with a bibliography. (If a student uses other people's results for his/her work, he/she has to name the source(s) of the results.)

Further details on the formal requirements for the thesis are specified in [Thesis sample](#).

ENCRYPTION OF THE THESIS

The thesis defence is public. Restrictions are only possible in the case of confidential theses as specified in Section 80 (ARS, section 79 (6))

The thesis may only contain information warranting its confidentiality in justified cases. If the thesis can be interpreted without the information warranting its confidentiality, (e.g. personal data can be anonymised), then there is no reason to include such information. (ARS, section 80 (2))

The thesis is to be classified as confidential (encrypted) if it

- a) contains business secrets,
- b) contains classified information,
- c) contains a patent or an invention,

- d) contains the personal data of a third party,
- e) concerns the University's business interests. (ARS, section 80 (4))

Documents required for requesting the encryption:

- **IK - Encryption of the thesis** in Neptun with the attachment of
- **Statement of consent** request form filled, signed and stamped by the head of the involved company or involved person. This document is required to be attached to the Neptun request in pdf format and available on the website of <http://csbsc.elte.hu/requests/>

The task of the supervisor is to assist the student in the elaboration of the thesis according to paragraph (2) and in the establishment of the justification of its classification as secret. The task of the student is to obtain the necessary permits and consents for the confidential thesis before submitting them. (ARS, section 80 (3))

The confidentiality of the thesis is to be announced by the Student or ordered by the supervisor or the chair of the final examination committee. (ARS, section 80 (5))

The confidential thesis shall be submitted with the protection provided by an electronic password, which may only be revealed to those entitled to the classified information in question. (ARS, section 80 (6))

Only the evaluation committee, the supervisor and the student, and in the case of a teacher training programme, the professional methodology instructor may be present at the defense of a confidential thesis. (ARS, section 80 (7))

Deadline for submitting the encryption documents:

- **1 November** in case of Final Exam in January
- **1 April** in case of Final Exam in June

After the approval of the Neptun request form by the vice-dean, the student will receive an encryption certificate. This has to be uploaded together with the thesis into Neptun. In case of encryption after the thesis is uploaded in Neptun, the „confidential” checkbox has to be ticked in.

REGISTRATION FOR FINAL EXAMINATION

The fact that examinees wish to take their Final Examination in the particular Final Examination period has to be registered by Education Office. (ARS, section 382, ad section 81 (6))

Registration for Final Examination can be submitted through Neptun:
Administration / Requests / Available request forms / IK – Registration for Final Examination (BSc)

Deadlines for submitting the Registration for Final Examination form:

- **1 November** in case of Final Exam in January
2-15 November: the application can be submitted with the payment of a late fee.
- **1 April** in case of Final Exam in June
2-15 April: the application can be submitted with the payment of a late fee.

Students are required to choose from the old and the new (2022) Final Exam Regulation on this form. In case of choosing the new regulation, it is mandatory to register 3 topics. Students have to know all the concepts and methods of the 3 chosen topics on the Final Examination.

SUBMITTING THE THESIS

Deadline for submitting thesis in Neptun:

- in case of taking the Final Examination in January: **1 December**
Extended deadline is not available.
- in case of taking the Final Examination in June 2025 and later: **1 May**
Extended deadline is not available.

Only one file can be uploaded in pdf or zip format. The uploaded file has to contain the following documents in addition to the thesis and the programme:

- [Cover](#)
- **Thesis Topic Registration Form**
- If needed: documents regarding [Encryption](#)

In case you can't print and sign the documents **listed above**, fill them in electronically and add the following abbreviation after your name: sgd.

For uploading data files bigger than 200 MB please follow these steps:

1. Upload in Neptun one zip file including the Thesis Topic Declaration, the Thesis, the source-code of your programme (concerning your own work), the Encryption request (if applicable) and a document where you indicate that the whole thesis programme is available on Onedrive and give access to the head of the Final Examination Committee. He/she will forward the documents towards the other members of the Committee.

2. Upload the whole thesis programme onto Onedrive, available via your inf mailing system.

Please note! The upload deadline applies for the documents uploaded on Onedrive (the date of upload is visible on Onedrive as well). We do not accept modifications or alterations after the deadline. It goes for the same for the documents uploaded in Neptun and for those files, what due to the size of the program were made available via a link.

Submitting Thesis in Neptun

After logging in Neptun click on Studies > Degree Thesis/Thesis application menu.

Here you can upload your thesis by clicking on „Upload degree thesis”.

Add the final title of the thesis, some keywords (3-5 words that will make searching easier) and the short summary of the thesis (max. half page).

After this you can upload and save the thesis itself in pdf or zip.

Save the file in the following name and format: NAME_NEPTUNCODE_THESIS.PDF OR .ZIP

Choose the correct file, wait until the upload is ready. (You will not receive a system message to wait.) Once the upload is ready, the name of the uploaded file will be listed.

Click on “Save file”.

Check that the system **has** really saved all the uploaded documents (via View degree thesis)

The Thesis file can be exchanged until the deadline for upload.

In case you would like to exchange the uploaded file: delete the first file, then upload the new, correct file.

In case of an encrypted thesis, don't forget to tick in „Confidential” option.

ASPECTS OF THE REVIEW

The student's thesis is reviewed by the supervisor and an external supervisor (if there is one). Theses are evaluated according to a set of criteria based on the following:

1. The difficulty of the programming task solved. Since the method and the algorithm are predetermined, their difficulty and complexity are not included in the evaluation.

2. Quality attributes of the program including: transparency, changeability, extensibility, adaptability to the user, reliability etc.
3. Usability of the User Documentation: How easily and quickly the user can get familiarized with the program?
4. In what extent does the Developer Documentation help orientation in the program, as well as in the further development and maintenance of the program?
The aspects considered:
 - the structure, language and appearance/layout of the documentation;
 - the transparency of the program text;
 - the relationship and harmony between the parts of the documentation (algorithms, modules and program structures);
 - quality assessment of testing.
5. Opinion on the student's work.

Detailed aspects of the review.

Thesis review form - for supervisors

Supervisor's deadline for evaluation:

- in case of Final Exam in January: **8 January**
- in case of Final Exam in June: **11 June**

PREREQUISITES FOR THE FINAL EXAMINATION

1. Registration for the Final Examination in Neptun.
 2. Before the Final Examination students have to obtain a pre-degree certificate (absolutory/ absolutorium)
 - a. Fulfil all the subjects of the curriculum
 - b. Fulfil the Thesis consultation subject
 - c. Fulfil internship (has to be administered in Neptun)
 3. Submit their thesis recommended by the supervisor for defence.
 4. Have no debts/liabilities in Neptun and towards the university/library
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OLD FINAL EXAMINATION REGULATION

1. The Final Examination consists of two parts: the defence of the thesis by the student and a professional examination provided the thesis defence was successful.
2. During the thesis defence, the candidate has 15 minutes to present the completed program and then he/she has to answer the committee's questions regarding the program presented. A brief presentation of the task and the methods used are welcome. It is the student's responsibility to prepare the appropriate environment and test cases to demonstrate the operation of the program.
3. During the Final Examination, the student is required to give an account of his/her professional knowledge and skills required for the chosen profession by discussing a topic from the pre-set Final Examination questions' list.
4. The Final Examination committee evaluates the thesis itself, its defence and the professional exam with a grade from 1 (fail) to 5 (excellent). If the thesis is evaluated as insufficient/failed (1), the student must rewrite his/her thesis/program. If either part of the Final Examination i.e. the defence of the thesis or the professional examination is evaluated as failed, the student may not retake it within the same exam period.
5. A thesis that wins top or first prize at the National Conference of Scientific Students' Associations ("OTDK") may be rewarded by the Dean with an excellent grade without any further evaluation if it complies with all the requirements for the thesis (§7 Section 79, ARS).
6. In accordance with Section 383/A of the ARS, the final grade of the Final Examination is the mathematical average of the grades received for the thesis, the defence of the thesis and the oral examination. The final grade of the degree certificate equals to the grade of the Final Examination.

NEW FINAL EXAMINATION REGULATION 2022 - TO EVERYONE'S ATTENTION

Due to the change in Sections 382 and 383/A of the ARS (ELTE Academic Regulations for Students), students of the BSc in Computer Science (all the curricula) may opt for the new regulations of the Final Examinations and the new grading system of the degree certificate. In this case, the Final Examination will consist of the defence of the thesis, which also includes an evaluation of the student's knowledge of the subject material related to the academic background of his/her thesis. The Final Examination Committee will ask questions of the 3 topics, previously chosen while on the Registration for the Final Examination form. Students has to know all the

concepts and methods of the 3 topics. The Student receives a combined grade for the thesis and the thesis defence, which shall be the grade received for the Final Examination. The grade of the degree certificate will be the average of the cumulative average of the subjects fulfilled during the programme (in Neptun: aggregated classic average) and the Final Examination grade.

In case of retaking an old regulation Final Examination, the new regulations cannot be chosen.